Homework 4

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Deterministic Noise

1.

To compute the deterministic noise, first we need to find

$$h^* = \operatorname*{argmin}_{h \in \mathcal{H}} E_{\text{out}}(h).$$

Suppose that $h(x) = w \cdot x$ and the probability density function of x is $p(x)$. Since x is uniformly sampled from $[0, 2]$, we have

$$\begin{aligned} E_{\text{out}}(h) &= \mathbb{E}_{x \sim P_{\text{data}}} \text{err}_{\text{SQR}}(h(x), f(x)) \\ &= \int_0^2 (wx - e^x)^2 \cdot p(x) dx \\ &= \int_0^2 (w^2 x^2 - w \cdot x e^x + e^{2x}) \cdot \frac{1}{2} dx \\ &= \frac{1}{2}(\frac{4w^2}{3} - e^2 w - w + \frac{e^4}{4} - \frac{1}{4}). \end{aligned} \tag{1}$$

By applying method of completing square, we can find that the smallest value of (1) occurs when $w = \frac{3+3e^2}{8}$. That is,

$$h^* = \frac{3 + 3e^2}{8} \cdot x.$$

Therefore, the deterministic noise for each x is

$$\left| e^x - (\frac{3 + 3e^2}{8}) x \right|.$$

Learning Curve

2.

The following proposition implies that the first statement is always false:

Proposition 2.1
Consider the 0/1 error, an arbitrary non-empty hypothesis set $\mathcal{H}$, and a learning algorithm $\mathcal{A}$ that returns one $h \in \mathcal{H}$ with the minimum E_{in} on any non-empty data set $\mathcal{D}$. That is,

$$\mathcal{A}(\mathcal{D}) = \operatorname*{argmin}_{h \in \mathcal{H}} E_{\text{in}}(h). \tag{2}$$

Assume that each example in $\mathcal{D}$ is generated i.i.d. from a distribution $\mathcal{P}$, and define $E_{\text{out}}(h)$ with respect to the distribution. Then we have

$$\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] \leq \mathbb{E}_{\mathcal{D}}[E_{\text{out}}(\mathcal{A}(\mathcal{D}))]. \tag{3}$$

Proof.
We prove this proposition by contradiction, suppose that

$$\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] > \mathbb{E}_{\mathcal{D}}[E_{\text{out}}(\mathcal{A}(\mathcal{D}))]. \tag{4}$$

Let $f(x)$ be the target function and

$$h^* = \operatorname*{argmin}_{h \in \mathcal{H}} E_{\text{out}}(h).$$

Since each example in $\mathcal{D}$ is generated i.i.d. from a distribution $\mathcal{P}$, for any $h \in \mathcal{H}$, we have

$$\begin{aligned}\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(h)] &= \mathbb{E}_{x\sim\mathcal{P}}[\text{err}_{0/1}(h(x), f(x))] \\ &= E_{\text{out}}(h).\end{aligned} \tag{5}$$

Because h^* has the smallest E_{out} over $\mathcal{H}$,

$$E_{\text{out}}(h^*) = \mathbb{E}_D[E_{\text{out}}(h^*)] \leq \mathbb{E}_D[E_{\text{out}}(\mathcal{A}(\mathcal{D}))]. \tag{6}$$

Similarly, by the definition of $\mathcal{A}(\mathcal{D})$,

$$\mathbb{E}_D[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] \leq \mathbb{E}_D[E_{\text{in}}(h^*)]. \tag{7}$$

Combining the results of (4) with (6) and (7) gives

$$\mathbb{E}_D[E_{\text{out}}(h^*)] \overset{(6)}{\leq} \mathbb{E}_D[E_{\text{out}}(\mathcal{A}(\mathcal{D}))] \overset{(4)}{<} \mathbb{E}_D[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] \leq \mathbb{E}_D[E_{\text{in}}(h^*)].$$

Hence, we have

$$E_{\text{out}}(h^*) = \mathbb{E}_D[E_{\text{out}}(h^*)] < \mathbb{E}_D[E_{\text{in}}(h^*)] \overset{(5)}{=} E_{\text{out}}(h^*).$$

This leads to a contradiction. Hence the statement must be true. □

To make sure that the other two statements are possibly true, we just need to find an example for each statement. For the second statement, consider that $f(x) = 1$ and $\mathcal{H} = \{h(x) \mid h(x) = [\![x = c]\!], c \in \mathbb{R}\}$. Therefore, given any hypothesis from $\mathcal{H}$, the value of $[\![h(x) = f(x)]\!]$ will be the same for all x and thus $\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] = \mathbb{E}_{\mathcal{D}}[E_{\text{out}}(\mathcal{A}(\mathcal{D}))]$. For the third statement, just consider the same hypothesis set but different target function $g(x) = x$ and $\mathcal{D}$ of size 1. Since there is only one data in $\mathcal{D}$, $\mathcal{A}$ can always find a hypothesis that perfectly fit the data set (i.e., $\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] = 0$). However, $\mathcal{A}(\mathcal{D})$ will make wrong prediction for all data points except the one in $\mathcal{D}$, which makes $\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(\mathcal{A}(\mathcal{D}))] < \mathbb{E}_{\mathcal{D}}[E_{\text{out}}(\mathcal{A}(\mathcal{D}))]$. In conclusion, both of the last two statements can possibly be true.

Noise Virtual Examples

3.

Let

$$X = \begin{bmatrix} \mathtt{x}_1^T \\ \vdots \\ \mathtt{x}_N^T \end{bmatrix}$$

and

$$\tilde{X} = \begin{bmatrix} \tilde{\mathtt{x}}_1^T \\ \vdots \\ \tilde{\mathtt{x}}_N^T \end{bmatrix}.$$

, where $\tilde{\mathtt{x}}_i = \mathtt{x}_i + \boldsymbol{\epsilon}_i$ and $\boldsymbol{\epsilon}_i$ is generated i.i.d. from multivariate normal distribution $\mathcal{N}(\mathbf{0}, \sigma^2 \cdot I_{d+1})$. Then, X_h can be expressed as

$$X_h = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix}. \tag{8}$$

Let $\epsilon_i^{(j)}$ denote the j-th term of $\boldsymbol{\epsilon}_i$, then we have

$$\tilde{X} = X + E. \tag{9}$$

, where E is a $N \times (d+1)$ matrix with $E_{i,j} = \epsilon_i^{(j)}$. Note that

$$\mathbb{E}(E) = O, \tag{10}$$

where O denotes a $N \times (d+1)$ zero matrix. Also note that

$$\mathbb{E}[\epsilon_i^{(j)} \cdot \epsilon_k^{(l)}] = \begin{cases} N\sigma^2, & \text{if } i = k \text{ and } j = l \\ 0, & \text{otherwise} \end{cases},$$

and thus

$$\mathbb{E}(E^T E) = N\sigma^2 I_{d+1}. \tag{11}$$

By block matrix multiplication,

$$X_h^T X_h = \begin{bmatrix} X^T \tilde{X}^T \end{bmatrix} \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} = X^T X + \tilde{X}^T \tilde{X}. \tag{12}$$

By (12), the expectation value of $X_h^T X_h$ is equal to

$$\begin{aligned}
\mathbb{E}[X_h^T X_h] &= \mathbb{E}[X^T X + \tilde{X}^T \tilde{X}] \\
&= X^T X + \mathbb{E}[X^T X] + \mathbb{E}[X^T E] + \mathbb{E}[E^T X] + \mathbb{E}[E^T E] \\
&\overset{(10)}{=} 2X^T X + O + O + \mathbb{E}[E^T E] \\
&\overset{(11)}{=} 2X^T X + N\sigma^2 I_{d+1}.
\end{aligned}$$

4.

Following the notation and results we have developed in **3.**, when we apply the block matrix multiplication, we will obtain

$$\mathbb{E}[X_h^T \boldsymbol{y}_h] = \mathbb{E}[X^T \boldsymbol{y} + \tilde{X}\boldsymbol{y}] = 2X^T \boldsymbol{y} + \mathbb{E}[E^T y] \overset{(10)}{=} 2X^T \boldsymbol{y}. \tag{13}$$

Regularization

5.

Let $\boldsymbol{w}_\lambda^*$ denotes the optimal solution of the optimization problem

$$\min_{\boldsymbol{w} \in \mathcal{R}^{d+1}} \frac{1}{N} \|Z\boldsymbol{w} - \boldsymbol{y}\|^2 + \frac{\lambda}{N} \boldsymbol{w}^T \boldsymbol{w}. \tag{14}$$

From the derivation of Lecture 14, we know that

$$\boldsymbol{w}_\lambda^* = (Z^T Z + \lambda I)^{-1} Z^T \boldsymbol{y}. \tag{15}$$

Since $Z = XQ$, we have

$$Z^T Z = Q^T X^T X Q = \underbrace{(Q^T Q)}_{I} \cdot \Gamma \cdot \underbrace{(Q^T Q)}_{I} = \Gamma. \tag{16}$$

Therefore,

$$\begin{aligned}
\boldsymbol{w}_\lambda^* &\overset{(16)}{=} (\Gamma + \lambda I)^{-1} Z^T \boldsymbol{y} \\
&= \begin{bmatrix} \frac{1}{\lambda+\gamma_0} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda+\gamma_1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda+\gamma_d} \end{bmatrix} \times Z^T \boldsymbol{y} \\
&= \begin{bmatrix} \frac{a_0}{\lambda+\gamma_0} \\ \vdots \\ \frac{a_d}{\lambda+\gamma_s} \end{bmatrix},
\end{aligned} \tag{17}$$

where a_i denotes the i-th entry of $Z^T \boldsymbol{y}$. So i-th entry of $\boldsymbol{v} = \boldsymbol{w}_\lambda^*$ and $\boldsymbol{u} = \boldsymbol{w}_0^*$ should be $a_i/(\lambda + \gamma_i)$ and a_i/γ_i, respectively. Hence, the ratio u_i/v_i should be

$$\frac{a_i}{\lambda + \gamma_i} \times \frac{\gamma_i}{a_i} = \frac{\gamma_i}{\lambda + \gamma_i}. \tag{18}$$

6.

We need to solve for the optimal solution for the following one-variable regularized linear regression problem:

$$\min_{w\in\mathbb{R}} \frac{1}{N}\sum_{n=1}^{N}(w\cdot x_n - y_n)^2 + \frac{\lambda}{N}w^2. \tag{19}$$

First we find the critical point of the objective function of the optimization problem above:

$$\begin{aligned}
0 &= \frac{d}{dw}\left(\frac{1}{N}\sum_{n=1}^{N}(w\cdot x_n - y_n)^2 + \frac{\lambda}{N}w^2\right)\Bigg|_{w=w^*} \\
&= \frac{1}{N}\cdot\frac{d}{dw}\left(\sum_{n=1}^{N}(x_n^2 w^2 - 2x_n y_n w + y_n^2) + \lambda w^2\right)\Bigg|_{w=w^*} \\
&= \frac{2}{N}\cdot\left(w\cdot(\sum_{n=1}^{N}x_n^2 + \lambda) - \sum_{n=1}^{N}x_n y_n\right).
\end{aligned} \tag{20}$$

Hence,

$$w^* = \frac{\sum_{n=1}^{N}x_n y_n}{\sum_{n=1}^{N}x_n^2 + \lambda}. \tag{21}$$

We can differentiate (20) to know that the second order derivative of the objective funciton is $\frac{2}{N}(\sum_{n=1}^{N}x_n^2+\lambda) > 0$ and thus it's convex. So w^* is the global minimizer of the regression problem (19). Since $C = (w^*)^2$, we get

$$C = (w^*)^2 = \left(\frac{\sum_{n=1}^{N}x_n y_n}{\sum_{n=1}^{N}x_n^2 + \lambda}\right)^2. \tag{22}$$

7.

First we compute the derivative of the objective function, denoted by $L(y)$, of the problem (with regularizer $\Omega(y)$):

$$\begin{aligned}
\frac{dL(y)}{dy} &= \frac{d}{dy}\left(\frac{1}{N}\sum_{n=1}^{N}(y-y_n)^2 + \frac{2K}{N}\Omega(y)\right) \\
&= \frac{1}{N}\sum_{n=1}^{N}(2y-2y_n) + \frac{2K}{N}\cdot\Omega'(y) \\
&= 2y - \frac{2}{N}\sum_{n=1}^{N}y_n + \frac{2K}{N}\cdot\Omega'(y).
\end{aligned} \tag{23}$$

Since all the options have the form $(y-d)^2$, where d is the unknown real number we are seeking for, (23) becomes

$$\begin{aligned}
\frac{dL(y)}{dy} &= 2y - \frac{2}{N}\sum_{n=1}^{N}y_n + \frac{2K}{N}\Omega'(y) \\
&= 2y - \frac{2}{N}\sum_{n=1}^{N}y_n + \frac{2K}{N}\frac{d}{dy}(y-d)^2 \\
&= 2y - \frac{2}{N}\sum_{n=1}^{N}y_n + \frac{2K}{N}(2y-2d).
\end{aligned} \tag{24}$$

If the estimate

$$y^* = \frac{\sum_{n=1}^{N}y_n + K}{N+2K}$$

is a minimizer of $L(y)$, then the value of derivative of $L(y)$ at this point must be zero. Thus, from (24) we have

$$
\begin{aligned}
& 0 = L'(y^*) \\
\Rightarrow\ & 0 = 2y^* - \frac{2}{N}\sum_{n=1}^{N} y_n + \frac{2K}{N}(2y^* - 2d) \\
\Rightarrow\ & \frac{2K}{N}\cdot d = \frac{\sum_{n=1}^{N} y_n + K}{N+2K} - \frac{1}{N}\sum_{n=1}^{N} y_n + \frac{2K\cdot\left(\sum_{n=1}^{N} y_n + K\right)}{N(N+2K)} \\
\Rightarrow\ & \frac{2K}{N}\cdot d = \frac{N\cdot(\sum_{n=1}^{N} y_n + K) - (N+2K)\cdot\sum_{n=1}^{N} y_n + 2K\cdot(\sum_{n=1}^{N} y_n + K)}{N(N+2K)} \\
\Rightarrow\ & \frac{2K}{N}\cdot d = \frac{\cancelto{0}{(N-(N+2K)+2K)}\cdot\sum_{n=1}^{N} y_n + K(N+2K)}{N(N+2K)} = \frac{K}{N} \\
\Rightarrow\ & d = 0.5.
\end{aligned}
$$

Therefore, $\Omega(y) = (y - 0.5)^2$.

8.

The linear regression problem in the $\mathcal{Z}$-space is

$$\min_{\tilde{\boldsymbol{w}}\in\mathcal{R}^{d+1}} \frac{1}{N}\sum_{n=1}^{N}(\tilde{\boldsymbol{w}}^T\Phi(\mathtt{x}_n) - y_n)^2 + \frac{\lambda}{N}(\tilde{\boldsymbol{w}}^T\tilde{\boldsymbol{w}}). \tag{25}$$

Let $\boldsymbol{w} = \Gamma^{-1}\tilde{\boldsymbol{w}}$. Thus,

$$\tilde{\boldsymbol{w}} = \Gamma\boldsymbol{w}. \tag{26}$$

Therefore, the optimization problem in (25) is equivalent to

$$\min_{\boldsymbol{w}\in\mathcal{R}^{d+1}} \frac{1}{N}\sum_{n=1}^{N}(\boldsymbol{w} - y_n)^2 + \frac{\lambda}{N}(\boldsymbol{w}^T\Gamma\Gamma\boldsymbol{w}). \tag{27}$$

So, $\Omega(\boldsymbol{w}) = \boldsymbol{w}^T\Gamma^2\boldsymbol{w}$.

9.

The scaled regularization problem is of the form

$$\min_{\boldsymbol{w}\in\mathbb{R}^{d+1}} \frac{1}{N}\sum_{n=1}^{N}(\boldsymbol{w}^T\mathtt{x}_n - y_n)^2 + \frac{\lambda}{N}\sum_{i=0}^{d}\beta_i w_i^2. \tag{28}$$

Note that

$$
\begin{aligned}
\lambda\cdot\sum_{i=0}^{d}\beta_i w_i^2 &= \lambda\left\|\begin{bmatrix}\sqrt{\beta}_0 w_0 & \cdots & \sqrt{\beta}_d w_d\end{bmatrix}^T\right\|^2 \\
&= \left\|\sqrt{\lambda}\begin{bmatrix}\sqrt{\beta}_0 & 0 & \cdots & 0 \\ 0 & \sqrt{\beta}_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\beta}_d\end{bmatrix}\begin{bmatrix}w_0 \\ w_1 \\ \vdots \\ w_d\end{bmatrix} - \begin{bmatrix}0 \\ 0 \\ \vdots \\ 0\end{bmatrix}\right\|^2.
\end{aligned} \tag{29}
$$

Therefore, let $\tilde{X} = \sqrt{\lambda}\sqrt{B}$ and $\boldsymbol{y} = \boldsymbol{0}$, then the optimization problem in (28) becomes

$$\min_{\boldsymbol{w}\in\mathbb{R}^{d+1}} \frac{1}{N}\sum_{n=1}^{N}(\boldsymbol{w}^T\mathtt{x}_n - y_n)^2 + \frac{1}{N}\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|^2, \tag{30}$$

of which the minimizer is the same as the minimizer of the following problem:

$$\min_{\boldsymbol{w}\in\mathbb{R}^{d+1}} \frac{1}{N+K}\left(\sum_{n=1}^{N}(\boldsymbol{w}^T\mathtt{x}_n - y_n)^2 + \sum_{i=0}^{d}(\boldsymbol{w}^T\tilde{\mathtt{x}} - \tilde{y}_n)^2\right). \tag{31}$$

Leave-one-out

10.

If a single instance is chosen as the validation data, then the number of instances of the same class will be $N-1$. Thus, $\mathcal{A}_{\text{majority}}$ will choose the constant classifier that always predicts the opposite class. As a result, the validation error will always be 1, so the leave-one-out error will be $\frac{1}{N} \cdot \sum_{i=1}^{N} 1 = 1$.

11.

Let the data set denoted by $\mathcal{D} = \mathcal{D}^+ \cup \mathcal{D}^-$, where the two non-empty sets D^+ and D^- are the set of instances labeled by $+1$ and the set of instances labeled by -1, respectively. Moreover, let $d_+ = (x_+, 1)$ denotes the instance whose features x_+ has the smallest value among all instances in $\mathcal{D}^+$. Similarly, let $d_- = (x_-, -1)$ denotes the instance whose features x_- has the largest value among all instances in $\mathcal{D}^+$. If the instance that is chosen as the validation set of size 1 is neither d_+ nor d_+, then the learning algorithm will return a decision stump $h_{+1,\theta}$ whose boundary θ lies between d_+ and d_- because such model has the lowest $E_{\text{in}}(h_{+1,\theta}) = 0$. Therefore, $h_{+1,\theta}$ will predict the label of the chosen validation instance correctly (i.e., $E_{\text{val}} = 0$). However, if the instance chosen as validation instance is d_+, then the learning algorithm will return a decision stump $h_{+1,\theta}$ whose boundary θ may be larger than x_+ but still has lowest $E_{\text{in}}(h_{+1,\theta}) = 0$. In this case, E_{val} may be 1. Similar situation may happen when d_- is chosen to be validation instance. Therefore, the worst case for leave-one-out error will be

$$E_{\text{loocv}}(\mathcal{H}, \mathcal{A}) = \frac{1}{N} \cdot (\underbrace{1}_{\mathcal{D}_{\text{val}}=\{d_+\}} + \underbrace{1}_{\mathcal{D}_{\text{val}}=\{d_-\}}) = \frac{2}{N}. \tag{32}$$

Thus, $\frac{2}{N}$ will be the tightest upper bound of leave-one-out error.

12.

For hypothesis of the form $h(x) = w_0$, if we choose (x_3, y_3) as the validation instance, then the best w_0 to minimize the training error $E_{\text{in}}(h) = \frac{1}{2}\left((w_0 - y_1)^2 + (w_0 - y_2)^2\right)$ would be $\frac{1}{2}(y_1 + y_2)$. Hence, $E_{\text{val}}(h)$ would be $(y_3 - \frac{1}{2}(y_1 + y_2))^2$. So the leave-one-out error is

$$\frac{1}{3}\left((y_3 - \frac{1}{2}(y_1 + y_2))^2 + (y_2 - \frac{1}{2}(y_1 + y_3))^2 + (y_1 - \frac{1}{2}(y_2 + y_3))^2\right) = \frac{1}{3}(1 + 4 + 1) = 2. \tag{33}$$

For hypothesis of the form $h(x) = w_0 + w_1 x$, if we choose (x_3, y_3) as the validation instance, then the best hypothesis to minimize the training error would be a line passing through the other two instances. That is,

$$h(x) = \frac{y_1 - y_2}{x_1 - x_2} \cdot x + \frac{x_1 y_2 - x_2 y_1}{x_1 - x_2}. \tag{34}$$

Thus, E_{val} is equal to

$$\left(y_3 - \left(\frac{y_1 - y_2}{x_1 - x_2} \cdot x_3 + \frac{x_1 y_2 - x_2 y_1}{x_1 - x_2}\right)\right)^2. \tag{35}$$

Use this formula, we can compute the leave-one-out error is

$$\frac{1}{3}\left(\left(-3 \cdot \frac{-2}{3-\rho} + \frac{6}{3-\rho} - 0\right)^2 + \left(3 \cdot \frac{2}{3+\rho} + \frac{6}{3+\rho} - 0\right)^2 + \left(\rho \cdot \frac{0}{3+3} + \frac{0}{3+3} - 2\right)^2\right). \tag{36}$$

Thus, if the two models are tied when using E_{looc}, (33) and (36) implies that

$$\frac{1}{3}\left(\left(-3 \cdot \frac{-2}{3-\rho} + \frac{6}{3-\rho} - 0\right)^2 + \left(3 \cdot \frac{2}{3+\rho} + \frac{6}{3+\rho} - 0\right)^2 + \left(\rho \cdot \frac{0}{3+3} + \frac{0}{3+3} - 2\right)^2\right) = 2. \tag{37}$$

Solving (37) for ρ gives

$$\rho = \pm\sqrt{81 + 36\sqrt{6}}. \tag{38}$$

13.

Since the validation examples are generated i.i.d. from the distribution $\mathcal{P}(\mathtt{x}, y)$, for any fixed hypothesis h, we have

$$\begin{aligned}
\underset{\mathcal{D}_{\text{val}}\sim\mathcal{P}^K}{\text{Variance}}[E_{\text{val}}(h)] &= \underset{\mathcal{D}_{\text{val}}\sim\mathcal{P}^K}{\text{Variance}} \frac{1}{K}\sum_{i=1}^{K}[\text{err}(h(\mathtt{x}_i), y_i)] \\
&= \frac{1}{K^2} \underset{\mathcal{D}_{\text{val}}\sim\mathcal{P}^K}{\text{Variance}} \sum_{i=1}^{K}[\text{err}(h(\mathtt{x}_i), y_i)] \\
&= \frac{1}{K^2}\sum_{i=1}^{K} \underset{(\mathtt{x},y)\sim\mathcal{P}}{\text{Variance}}[\text{err}(h(\mathtt{x}), y)] \qquad (39) \\
&= \frac{1}{K} \underset{(\mathtt{x},y)\sim\mathcal{P}}{\text{Variance}}[\text{err}(h(\mathtt{x}), y)]. \qquad (40)
\end{aligned}$$

(39) follows since the variance of sum of independent variables equals to the sum of their individual variances.

Learning Principles

14.

Given 4 vertices of a rectangle in $\mathbb{R}^2$ as input vectors, we can use the 2D perceptrons to implement all dichotomies but two special cases. Here comes the prove:

Proposition 14.1
Given 4 vertices of a rectangle in $\mathbb{R}^2$ as input vectors $\mathtt{x}_1$, $\mathtt{x}_2$, $\mathtt{x}_3$ and $\mathtt{x}_4$, with $\mathtt{x}_1$ and $\mathtt{x}_3$ on the same diagonal of the rectangle (i.e., $\mathtt{x}_2$ and $\mathtt{x}_4$ are on the other diagonal of the rectangular). If the labels of $\mathtt{x}_1$ and $\mathtt{x}_3$ are the same but different from the labels of $\mathtt{x}_2$ and $\mathtt{x}_4$ (this implies $\mathtt{x}_2$ and $\mathtt{x}_4$ have the same label value), then there is no 2D perceptron that can implement this dichotomy.

Proof.
If there exist a 2D perceptron $h(\mathtt{x}) = \text{sign}(\boldsymbol{w}\mathtt{x})$ that implements this dichotomy, then we have

$$\text{sign}(\boldsymbol{w}\mathtt{x}_1) = \text{sign}(\boldsymbol{w}\mathtt{x}_3) = \text{sign}(\boldsymbol{w}(\mathtt{x}_1 + \mathtt{x}_3)) \neq \text{sign}(\boldsymbol{w}\mathtt{x}_2) = \text{sign}(\boldsymbol{w}\mathtt{x}_4) = \text{sign}(\boldsymbol{w}(\mathtt{x}_2 + \mathtt{x}_4)). \qquad (41)$$

Moreover, since the inputs form a rectangular, we have

$$\mathtt{x}_1 + \mathtt{x}_3 = \mathtt{x}_2 + \mathtt{x}_4.$$

Therefore,

$$\text{sign}(\boldsymbol{w}(\mathtt{x}_1 + \mathtt{x}_3)) = \text{sign}(\boldsymbol{w}(\mathtt{x}_2 + \mathtt{x}_4)), \qquad (42)$$

which contradicts to (41). Thus, such perceptron dosn't exist. □

For the dichotomies described in the proposition above, the smallest value of E_{in} is $1/4$ since we must sacrifice one instance to make the perceptron make correct predictions for the others. For the other dichotomies, it can be easily shown by visualization that each dichotomy can be implemented by a certain 2D perceptron. Therefore, their smallest values of E_{in} are all 0. From the discussion above, we have

$$\mathbb{E}_{y_1,y_2,y_3,y_4}\left(\min_{\boldsymbol{w}\in\mathbb{R}^{2+1}} E_{\text{in}}(\boldsymbol{w})\right) = 2 \times \frac{1}{4} \times \frac{1}{16} = \frac{2}{64}. \qquad (43)$$

15.

If $P(y = +1) = p$, then $E_{\text{out}}(g)$ should be

$$P(y = +1) \cdot P(g(\mathtt{x}) = -1 \mid y = +1) + P(y = -1) \cdot P(g(\mathtt{x}) = +1 \mid y = -1) = p \cdot \epsilon_+ + (1 - p) \cdot \epsilon_-. \qquad (44)$$

When g is as good as (or as bad as) g_c, $E_{\text{out}}(g) = E_{\text{out}}(g_c) = 1 - p$. Therefore, we have

$$\begin{aligned}
p \cdot \epsilon_+ + (1 - p) \cdot \epsilon_- &= 1 - p \qquad (45) \\
\Rightarrow p &= \frac{1 - \epsilon_-}{\epsilon_+ - \epsilon_- + 1}. \qquad (46)
\end{aligned}$$